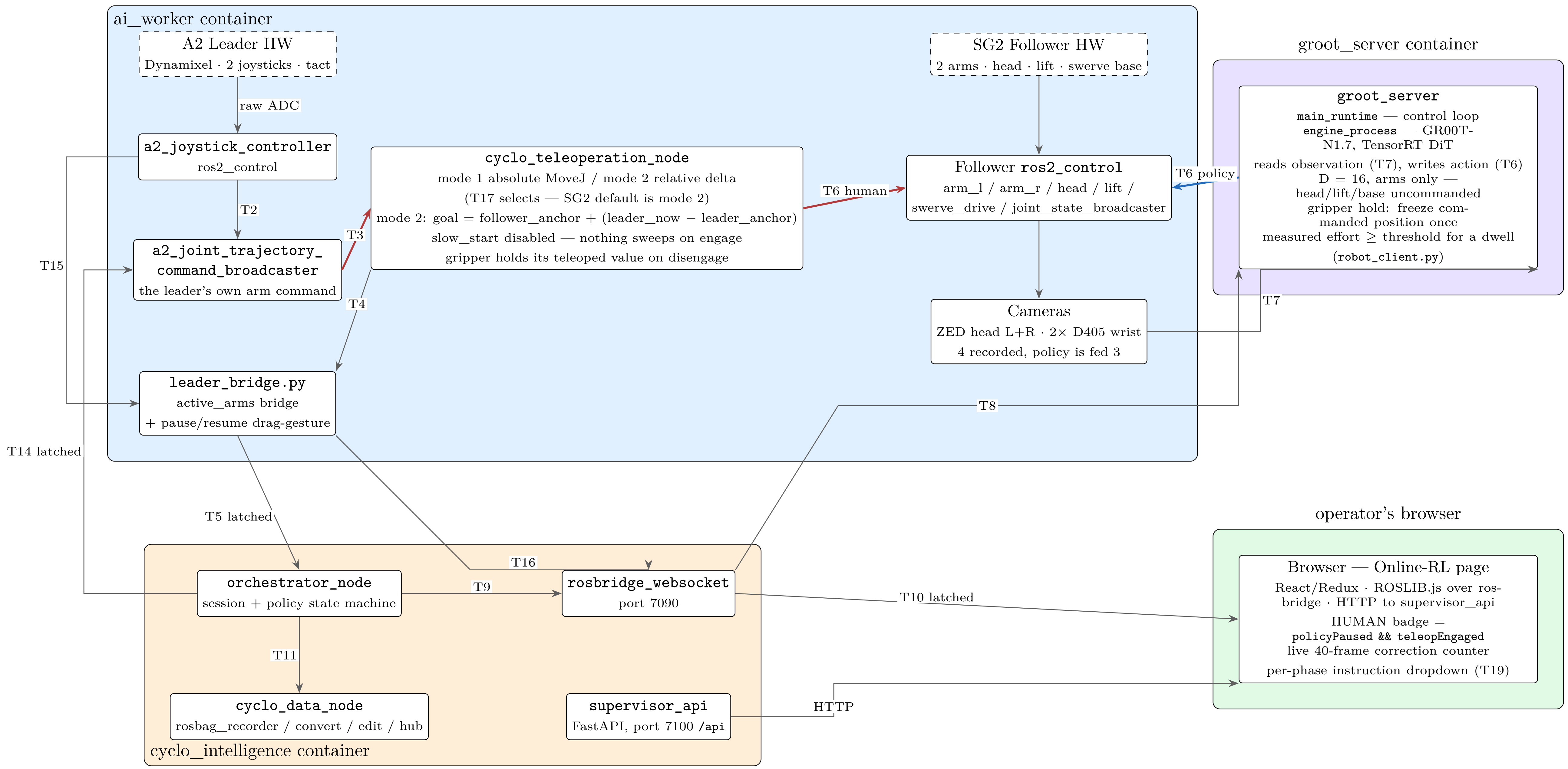
HG-DAgger / Online-RL — Runtime Control Path

FFW SG2 follower · A2 leader · GR00T-N1.7 policy

human commands the arms   policy commands the arms   status / feedback / control plane.   Edge labels are IDs defined in the legend on p.3.



The operator's takeover loop

1. joystick drag-down ≥1 s → T16 → INFERENCING → PAUSED
2. tact press → T2 `COMMAND_TOGGLE` → that arm engages
3. tact press again → arm disengages, gripper holds its teleoped value
4. joystick drag-down ≥1 s → T16 → PAUSED → INFERENCING

The gaps 1→2 and 3→4 are PAUSED with no arm engaged: labelled −1, not human.

T16 is one topic with two meanings, resolved by the phase it arrives in.

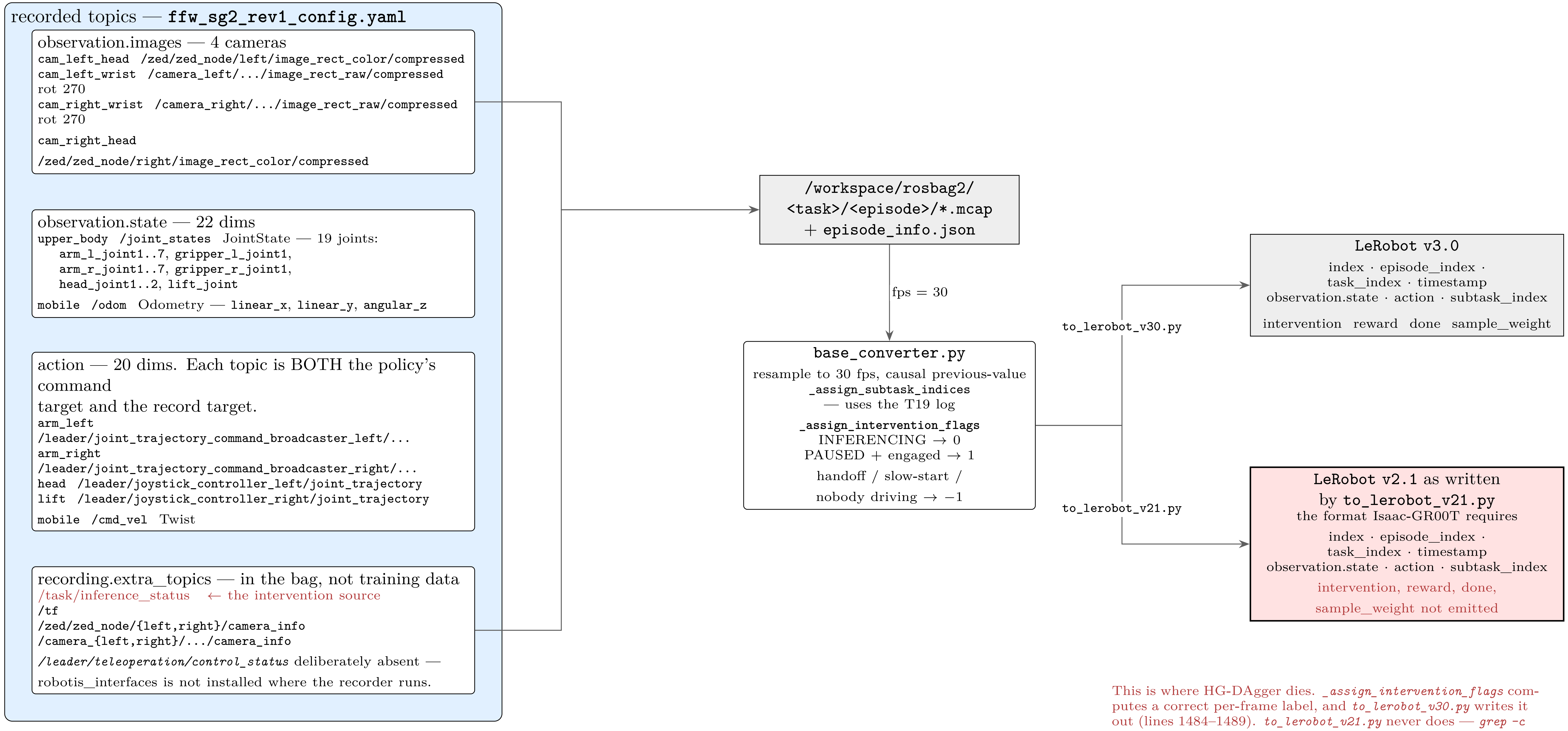
T6 is ONE topic, written by BOTH sides *`/leader/joint_trajectory_command_broadcaster_{left,right}/joint_trajectory`*

Teleop node and policy publish to the same place, so *`action`* is valid in either mode — no stale-leader pose. T14 stops them contending.

T18 Home goes straight from the browser to *`ai_worker`* over rosbridge — the only UI action that skips the orchestrator.

Recording and Conversion — what survives to training

Every recorded topic, and where the online-RL labels are lost



This is where HG-DAgger dies. *_assign_intervention_flags* computes a correct per-frame label, and *to_lerobot_v30.py* writes it out (lines 1484–1489). *to_lerobot_v21.py* never does — *grep -c intervention* on it returns 0. Isaac requires v2.1, so every GR00T export silently loses the human/policy label and degrades to plain behaviour cloning. Fix: *IMPROVEMENTS.md* §3.

Topic / Service Legend

Verified against `cyclo_intelligence_backup@main` and `aiw_backup@feature-a2`, 2026-08-30.

ID	Kind	Name	Type	Meaning / gotcha
T1	PUB/SUB	`/leader/joystick_controller/tact_trigger`	`std_msgs/String`	Tap emits ”left”/”right”. Record trigger disabled — one press also engages that arm, which during INFERENCING would put the leader onto T6 against the policy.
T2	PUB/SUB	`.../joystick_command`	`robotis_interfaces/ TeleoperationCommand`	ENABLE / DISABLE / TOGGLE, per arm. The tact sends TOGGLE — engage and stop are the same press.
T3	PUB/SUB	`.../raw_joint_trajectory`	`trajectory_msgs/ JointTrajectory`	The leader's own arm command, before mode handling.
T4	PUB/SUB	`/leader/teleoperation/control_status`	`robotis_interfaces/ ControlModeStatus`	`active_arms` + mode. Use T5.
T5	PUB/SUB latched	`/leader/teleoperation/active_arms_str`	`std_msgs/String`	Same information as T4 as a plain string, from `leader_bridge.py`. The UI badge reads it; the bag should too.
T6	PUB/SUB	`/leader/joint_trajectory_command_ broadcaster_{left,right}/joint_trajectory`	`trajectory_msgs/ JointTrajectory`	Written by BOTH the teleop node and the policy. That is why `action` is valid in either mode. T14 prevents contention, not separate topics.
T7	PUB/SUB	`/joint_states`, `/odom`, 4 camera topics	`JointState, Odometry, CompressedImage`	Follower feedback and vision. 4 cameras recorded, 3 declared to the policy.
T8	PUB/SUB	/<backend>/engine_command	`std_msgs/String`	Orchestrator → `groot_server`: load / unload / run.
T9	SERVICE	`/task/command`	`SendCommand.srv`	Start / Pause / Resume / Clear policy; Start / Stop / Cancel recording.
T10	PUB/SUB latched	`/task/inference_status`	`interfaces/InferenceStatus`	`inference_phase`: READY 0, LOADING 1, INFERENCING 2, PAUSED 3. Transitions only — the converter forward-fills. Sole source of the intervention label.
T11	SERVICE	`/data/recording`	`RecordingCommand.srv`	Recording lifecycle, orchestrator → `cyclo_data_node`.
T13	SERVICE	`/data/convert`, `/data/convert/status`	`StartConversion.srv, GetConversionStatus.srv`	LeRobot conversion. Both `convert_v21` and `convert_v30` false means ”run both”.
T14	PUB/SUB latched	`/task/policy_active`	`std_msgs/Bool`	Mirrors T10's INFERENCING into `ai_worker`. Gates the leader chain off T6 so the two never publish together.
T15	PUB/SUB	`.../sensorxel_joy_values`	`std_msgs/Float64MultiArray`	Raw joystick axes. The gesture detector watches these.
T16	PUB/SUB	`/leader/teleoperation/toggle_pause_request`	`std_msgs/Empty`	Fires once on a $\geq$1 s drag-hold. One topic, two meanings: pause if INFERENCING, resume if PAUSED. Lets the operator work without the browser.
T17	PUB/SUB	`.../control_command`	`robotis_interfaces/ ControlModeCommand`	Mode 1 absolute MoveJ, mode 2 relative delta. SG2 default is mode 2.
T18	PUB/SUB latched	`/task/home_in_progress`	`std_msgs/Bool`	One-shot Home trajectory, published straight from the browser — the only UI action that skips the orchestrator.
T19	SERVICE	`/task/command` (UPDATE_INSTRUCTION)	`SendCommand.srv`	Appends to `instruction_change_log`, which splits the episode into correctly-timed sub-segments. Without it a continuous take gets one segment and every frame carries the last instruction set.

Derived dataset columns

- `observation.state` (22) and `action` (20) — they differ: state carries `head_joint1/2`, action does not. Slice order follows YAML insertion order. The policy declares only the first 16 (both arms); the remainder are recorded but not trained on.
- `task_index` / `subtask_index` — per frame, from `episode_info.json`'s `segments`, split by `instruction_change_log` (T19). The episode's top-level `task_instruction` is a separate stable field; it is not the live per-phase instruction, which is overwritten on every T19 click.
- `intervention` — 1 = human drove, 0 = policy drove, -1 = excluded (handoff, nobody driving, phase never observed — note the handoff rule assumes a mode-1 sweep, see `IMPROVEMENTS.md` Defect 2). This is the inverse of `task_is_policy`, the name some downstream tooling expects; getting the polarity wrong raises nothing and trains on exactly the wrong frames. v3.0 only.
- `reward`, `done`, `sample_weight` — emitted alongside `intervention`. v3.0 only, so HIL-SERL-style training is also impossible from a v2.1 export.